Practical 2: Read Data from an LDR Sensor and Automatically Control an LED Based on Light Intensity Using Arduino

A) Objective

- To understand the working principle of a Light Dependent Resistor (LDR) and how it functions as an analog sensor.
- To build a voltage divider circuit using an LDR and a fixed resistor, and read the analog output using Arduino's analogRead() function.
- To implement threshold-based automatic LED control: the LED turns ON when ambient light falls below a set threshold (dark condition) and turns OFF when light is sufficient (bright condition).
- To develop an understanding of analog signal reading, value mapping, and conditional logic in Arduino programming.

B) Practical Outcome

On successful completion of this practical, the student will be able to:

- Explain the relationship between light intensity and LDR resistance, and how this is converted to a voltage reading via a voltage divider.
- Connect an LDR-based voltage divider circuit to an Arduino analog input pin.
- Write an Arduino sketch that reads analog sensor values and uses conditional logic to control a digital output (LED).
- Calibrate the threshold value by observing Serial Monitor readings under varying light conditions.

C) Expected Affective Domain

(Attitudes, safety habits, and professional practices this practical aims to instil)

- Develops analytical thinking: students learn to first observe raw sensor values before writing control logic, rather than hardcoding threshold guesses without data.
- Instils calibration awareness — real IoT sensors require characterization under actual deployment conditions; students experience this by testing under torch light vs. room light vs. covered sensor.
- Encourages iterative design: students adjust the threshold value in code based on observed data, reinforcing that engineering solutions are refined through testing, not fixed on first try.
- Builds appreciation for analog sensor signal behaviour — values drift, vary by unit, and depend on environmental context — preparing students for more complex sensor-fusion work later in the course.

D) Resources Required

Category	Items
Hardware	Arduino Uno board, USB Type-B cable, breadboard, 1 LDR (photoresistor), 1 resistor (10kΩ), 1 LED (any colour), 1 resistor (220Ω for LED), jumper wires (male-male)
Software	Arduino IDE (v2.x) with Serial Monitor enabled
Computer	PC/laptop with Arduino IDE installed and COM port access
Reference	Arduino Uno analog pin map (A0–A5); LDR datasheet

E) Experimental Setup and Code

Circuit Description

The LDR and a 10kΩ fixed resistor form a voltage divider connected between 5V and GND. The midpoint voltage — which changes with light intensity — is read by Arduino Analog pin A0. Separately, an LED with a 220Ω series resistor is connected to digital pin 8.

Working principle: In bright light, LDR resistance drops → midpoint voltage rises → analogRead() returns a higher value. In darkness, LDR resistance rises → midpoint voltage drops → analogRead() returns a lower value. The circuit orientation below assumes LDR is on the VCC side and fixed resistor is on the GND side.

Wiring Table

Component Pin	Connects To (Board Pin)	Notes
LDR Terminal 1	5V (Arduino)	LDR is not polarised; either terminal may connect to 5V
LDR Terminal 2	Analog Pin A0 AND one end of 10kΩ resistor	Junction point (voltage divider midpoint)
10kΩ Resistor other end	GND (Arduino)	Completes the voltage divider to ground
LED Anode (long leg)	Digital Pin 8 (via 220Ω resistor)	Current-limiting resistor in series
LED Cathode (short leg)	GND (Arduino)	Completes the LED circuit

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//Draw the Circuit diagram to interface the LDR with Arduino
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Part 1: Read and Observe Raw LDR Values

Before writing control logic, first read and record LDR values across different lighting conditions so the threshold can be set meaningfully.

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//Write the code to read the LDR value and print on Serial Monitor
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- Upload this sketch and open the Serial Monitor (Tools → Serial Monitor, baud rate 9600).
- Record the analogRead() value under three conditions: (1) normal room light, (2) torch/phone flashlight aimed at LDR, (3) finger/hand fully covering the LDR (simulating darkness).
- Use your recorded values to decide a threshold: a value between the 'bright' and 'dark' readings that separates the two states.

Observation Table — Fill During Part 1

Light Condition	Observed analogRead() Value	LED State Expected [Bright/Dark/ Medium]
Normal room light		
Bright torch on LDR		
Hand fully covering LDR		
Chosen threshold value		Between bright and dark readings

Part 2: Automatic LED Control Based on Threshold

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// Code for Automatic LED control based on LDR light intensity
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- Replace the threshold value (500) with the value you determined from your Part 1 observations.
- Upload and test: cover the LDR with your hand — the LED should turn ON. Remove your hand or shine a torch on it — the LED should turn OFF.
- If the LED does not respond as expected, adjust the threshold up or down in the code and re-upload based on your recorded readings.
- Note: analogRead() returns values in the range 0–1023 (10-bit ADC). 0 = 0V, 1023 = 5V.

F) Outcome / Observation

Record the following results in your lab journal after completing both parts:

Observation Point	Expected / Student-Recorded Result
analogRead() value in bright torch light	
analogRead() value when LDR fully covered	
Threshold value chosen by student	
LED state when LDR is covered (dark)	
LED state when torch shines on LDR	
Serial Monitor output correctness	

Students should record the specific threshold value they chose and briefly justify why that value was appropriate for their measurement environment. If the threshold needed to be adjusted after initial testing, note what changed and why.

G) MCQs for the Practical

Q1. What is the resistance behaviour of an LDR when light intensity increases?

A) Resistance increases

B) Resistance decreases

C) Resistance remains constant

D) Resistance becomes zero

Q2. In the voltage divider circuit (LDR on VCC side, 10kΩ resistor on GND side), what happens to the analog pin voltage when the LDR is covered (dark)?

A) Voltage increases

B) Voltage decreases

C) Voltage stays at 5V

D) Voltage stays at 0V

Q3. Which Arduino function is used to read the analog voltage from the LDR circuit?

A) digitalRead()

B) analogWrite()

C) analogRead()

D) Serial.read()

Q4. What is the range of values returned by analogRead() on an Arduino Uno?

A) 0 to 255

B) 0 to 512

C) 0 to 1023

D) 0 to 5000

Q5. Why is a 10kΩ fixed resistor used in series with the LDR?

A) To amplify the LDR signal

B) To form a voltage divider so the changing LDR resistance produces a changing voltage readable by the analog pin

C) To protect the LDR from overcurrent

D) To convert the LDR's analog output to a digital signal

Q6. In the Part 2 sketch, what will happen if the threshold value is set too high (e.g., 1000)?

A) The LED will never turn ON because the LDR value rarely exceeds 1000

B) The LED will always be ON regardless of light

C) The LED will blink very rapidly

D) The Arduino will reset automatically

Q7. Why does the student need to observe raw analogRead() values (Part 1) before writing the threshold logic (Part 2)?

A) To test if the Serial Monitor is working

B) To calibrate the threshold to the actual sensor's range under real ambient conditions

C) To verify the LED wiring is correct
D) To check the baud rate setting

Q8. Which real-world IoT application uses the same light-sensing and automatic switching principle as this practical?

A) Automatic street light control
B) Heart rate monitoring
C) Gas leak detection
D) Soil moisture-based irrigation

Q9. If the LDR and the 10kΩ resistor positions in the voltage divider are swapped (resistor on VCC side, LDR on GND side), what changes?

A) The circuit stops working entirely
B) The analogRead() value becomes higher in darkness instead of lower
C) The LED no longer lights up
D) The Serial Monitor stops printing

Q10. What does the Serial.println() function output compared to Serial.print()?

A) They are identical in output
B) Serial.println() adds a newline character at the end, moving the cursor to the next line
C) Serial.println() outputs to a file instead of the monitor
D) Serial.println() only works inside setup()

H) Assessment Rubrics

Total marks suggested: 10 (adjust to institutional scheme as needed)

Assessment Criteria	Max Marks
Pre-lab Preparation	_____ /5
Hardware setup & interfacing	_____ /6
Implementation & Execution/Output	_____ /6
Viva-voce	_____ /3
Journal/record	_____ /5
Total	_____ /25

Sign with Date

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